An Integrated AI, IoT and GIS-Based Platform for Flood-Risk Prediction, Real-Time River Monitoring, Post-Flood Disease Screening and Relief Distribution Management System

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Abstract*—Flood management systems often treat prediction, environmental monitoring, healthcare response, rescue coordination, and relief distribution as separate activities, limiting coordinated decision-making during flood events. This study proposes an integrated flood-disaster management platform combining Artificial Intelligence (AI), Internet of Things (IoT), and Geographic Information System (GIS) technologies within a single decision-support environment. The platform consists of four interconnected modules: (1) machine-learning-based flood-risk prediction with interactive GIS visualization; (2) IoT-enabled multi-point ultrasonic drainage monitoring for blockage detection and short-term water-level forecasting; (3) AI-assisted post-flood health-risk screening combining reported symptoms with image analysis; and (4) Multi-Output Random Forest-based rescue and relief-resource prioritization. All modules exchange information through a centralized integration layer, enabling flood-risk, river-condition, health, rescue, and relief data to support one another in near real time. The proposed approach aims to improve coordination, early response, healthcare prioritization, and humanitarian resource allocation by transforming fragmented flood-management activities into a unified, data-driven operational workflow.

Keywords*—Flood prediction, Geographic Information System, Internet of Things, Artificial Intelligence, machine learning, disaster management, post-flood health screening, rescue operations, relief distribution.

I. INTRODUCTION

Flooding remains a major natural hazard in Sri Lanka, particularly during monsoon periods, when intense rainfall causes widespread inundation, property damage, displacement, and disruption of essential public services [1]. Flood events also create public-health challenges, as contaminated water and poor sanitation increase the likelihood of water-borne and communicable diseases in affected communities [2], [3]. Managing these impacts effectively requires more than flood forecasting alone — it requires coordinated environmental monitoring, healthcare prioritization, rescue operations, and humanitarian relief distribution that follow recognized humanitarian principles concerning the adequacy, timeliness, and quality of assistance [4]. In practice, however, flood-risk information, river-monitoring data, health-related records, rescue requests, and relief-resource records are often handled independently by different operational teams, limiting information sharing and reducing decision-making efficiency.

Machine learning has proven effective for flood-risk prediction and susceptibility mapping [5], [6], IoT-based sensing has been widely explored for continuous water-level and flood monitoring [7], [8], AI has been investigated for post-disaster health surveillance [9], [10], and optimization approaches have been applied to humanitarian resource allocation [11]. However, relatively limited attention has been given to integrating flood prediction, river-condition monitoring, post-flood healthcare support, rescue coordination, and relief planning into a single data-sharing platform — the primary motivation for this study.

This research introduces an integrated flood-disaster management platform composed of four interconnected modules: GIS-based flood-risk prediction using ensemble machine learning [5], [6]; real-time drainage monitoring using multi-point IoT ultrasonic sensor nodes for blockage detection [7], [8]; AI-assisted post-flood health-risk

screening combining symptoms and images [9], [10]; and Multi-Output Random Forest-based rescue and relief-resource prioritization guided by humanitarian standards [5], [11], [4]. All four modules share a common integration layer so that information generated by one module — such as a drainage-blockage alert or an updated flood-risk classification — can immediately support decisions made in another, rather than remaining confined to a single tool. The main contributions of this work are:

- A review of existing work on machine-learning-based flood prediction, IoT river monitoring, post-flood health-risk assessment, and humanitarian resource allocation, highlighting the limited integration between these areas.
- A unified four-module architecture that exchanges flood-risk, river-monitoring, healthcare, rescue, and relief information through a common integration layer.
- A module-specific methodology combining GIS-based flood-risk classification, multi-point ultrasonic drainage monitoring, AI-assisted health-risk screening, and Multi-Output Random Forest relief prioritization, together with a per-module evaluation framework.

The remainder of the paper reviews related work (Section II), describes the proposed methodology (Section III) and system architecture (Section IV), summarizes the implementation (Section V), presents results and discussion (Section VI), and concludes with future directions (Section VII).

II. RELATED WORK

A. GIS-Based Machine Learning for Flood-Risk Prediction

Random Forest is widely adopted for flood-risk and susceptibility assessment due to its ability to model non-linear relationships among rainfall, elevation, slope, drainage, and land-use variables [5]. Demissie et al. showed that combining machine learning with GIS data improves flood-susceptibility mapping [6]. However, most such studies generate static or periodically updated maps with limited integration of continuously changing rainfall and river-level observations. Module 1 addresses this gap by connecting machine-learning predictions to an interactive GIS interface that updates as new monitoring data arrive.

B. IoT-Based River and Water-Level Monitoring

IoT technologies enable continuous, low-cost environmental monitoring. Moreno et al. developed RiverCore, a low-cost system for monitoring river levels over cellular networks [7], and Arshad et al. reviewed IoT and computer-vision techniques for flood monitoring [8]. Many implementations, however, rely on single-point measurements or fixed thresholds, which cannot always distinguish an obstruction from normal rainfall-driven variation. Module 2 instead compares water levels across multiple upstream–downstream points, using relative differences rather than an isolated threshold to detect flow abnormalities.

C. Post-Flood Disease Surveillance and Healthcare Decision Support

Machine learning has been applied to post-disaster health surveillance: Safari et al. compared several models for predicting infectious-disease incidence after flooding, with Random Forest achieving an AUC of 0.76 [9]. U-Net [10] remains a foundational architecture for biomedical image segmentation that can support image-based screening. These techniques are typically studied in isolation and oriented toward retrospective, population-level surveillance rather than rapid, individual-level prioritization — the gap addressed by Module 3.

D. Disaster-Relief Resource Allocation and Distribution Optimization

Huang et al. proposed a multi-objective emergency-response model balancing effectiveness and fairness [11], and Sakiani et al. proposed a dynamic inventory-routing model for redistributing relief goods [12]. The Sphere Handbook defines internationally recognized minimum humanitarian standards [4]. These approaches typically treat resource allocation as an isolated logistics problem rather than one informed by real-time flood, accessibility, and health data — a limitation addressed in Module 4 through a Multi-Output Random Forest model that references humanitarian standards [5], [11], [4].

Overall, effective technical approaches already exist for each of these functions individually. The gap addressed in this study is therefore not the absence of individual technologies, but the limited integration of these capabilities within a shared flood-disaster management platform. The proposed system responds to this gap by connecting the four functions through a common data-sharing and decision-support architecture, so that information generated by one function is available to inform decisions in another rather than remaining confined within a single tool.

III. METHODOLOGY

The proposed platform comprises four functional modules connected through a common integration layer. Although each module performs a distinct operational task, relevant outputs are exchanged between modules to support coordinated decision-making.

A. GIS-Based Flood Prediction and Mapping

Module 1 estimates flood risk at different locations using historical flood records, rainfall, river-water level, elevation, land-use, drainage, and soil variables, consistent with commonly used flood-conditioning factors [6]. Random Forest and Gradient Boosting classifiers are evaluated for risk estimation, with Logistic Regression retained as a baseline [5], [6], [13]. Each location is classified as Low, Moderate,

High, or Critical risk; predictions are converted into a GeoJSON layer and displayed on a Leaflet-based interactive map that updates as new rainfall or river-level data arrive, in contrast to a static susceptibility map. Module 1 also transmits flood-risk outputs to other modules through the integration layer.

B. IoT-Based Drainage Monitoring and Blockage Detection

Module 2 continuously monitors drainage conditions using IoT sensor nodes combining ultrasonic water-level sensors, a rainfall sensor, and a Wi-Fi-connected microcontroller, following established low-cost remote-monitoring approaches [7], [8]. Three ultrasonic sensors are positioned at upstream, middle, and downstream points; the water-level difference between neighbouring points is computed as

$$\textit{Water-Level Difference} = WL(\textit{Upstream}) - WL(\textit{Neighbouring}) \quad (1)$$

Rather than alerting from a single measurement, the module evaluates the magnitude and persistence of this difference: a large, persistent deviation indicates a possible blockage and its approximate location, while repeated normal differences indicate normal flow. Rainfall variation, tributary inflow, and known reservoir activity are considered to reduce false alerts. Detected conditions are classified into five alert levels (Table I). Module 2 also forecasts near-term water levels and forwards alerts to Module 1 (risk-map refresh) and Module 4 (accessibility constraints for rescue-route planning).

Alert Level	Trigger Condition
Normal	Difference within normal range
Low-risk anomaly	Minor deviation explained by rainfall/tributary
Medium-risk possible blockage	Deviation exceeds threshold briefly
High-risk possible blockage	Deviation persists beyond travel time
Critical blockage warning	Sustained high deviation; verification requested

TABLE I. DRAINAGE-CONDITION ALERT LEVELS (MODULE 2)

C. AI-Powered Post-Flood Disease Detection and Healthcare Monitoring

Module 3 supports early identification and prioritization of post-flood health concerns. Users submit symptom-related information — age group, symptoms, duration — with an optional image for additional visual evidence, building on prior evidence of machine-learning-based post-flood disease-risk prediction [9]; image preprocessing may use U-Net-based segmentation where required [10]. A hybrid approach combines machine-learning outputs with predefined risk rules to flag symptom combinations warranting attention even under limited model confidence, categorizing cases as Low, Moderate, High, or Critical/Urgent. High- and Critical/Urgent cases are highlighted on the healthcare dashboard for prioritized review; the module supports screening rather than independent diagnosis.

D. Post-Flood Rescue and Ration Distribution Management

Module 4 integrates rescue coordination with relief-camp prioritization. Public users submit rescue, food, water, medical, and shelter requests; officers assign rescue teams and manage distribution. Each camp record includes population, vulnerable-group counts, available food/water/medicine/sanitary stock, capacity, accessibility, distance from distribution points, and time since last delivery. Because several priorities must be predicted from one record, a Multi-Output Random Forest model estimates overall camp priority together with resource-specific priorities for food, water, medicine, and sanitary supplies [5]. Predictions are reviewed against humanitarian urgency, fairness, and Sphere Handbook minimum standards [11], [4] before officers finalize allocation decisions. Accessibility data from Module 2 flags unsafe routes before rescue or delivery teams are dispatched, and delivery outcomes are logged for future evaluation.

IV. SYSTEM ARCHITECTURE

Fig. 1 shows the overall system architecture. Each module ingests its own data — rainfall/water-level feeds, IoT telemetry, symptom/image submissions, and relief-camp/rescue-request data — and publishes processed outputs to a centralized integration layer built on REST APIs, a WebSocket/Socket.IO real-time bus, and a shared MongoDB store. Disaster-management officers, the public-facing application, and the rescue-and-relief operation centre all read from and write to this shared layer, so that a Module 2 blockage alert is immediately visible on the Module 1 dashboard, and Module 2 accessibility status can inform Module 4 rescue-route decisions.

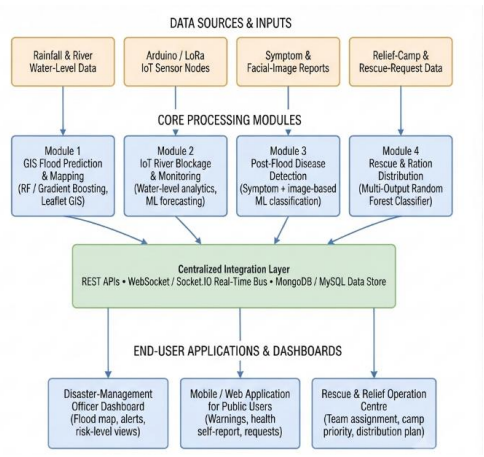


Fig. 1. Overall system architecture of the integrated platform.

Fig. 2 summarizes how the four modules cooperate during an actual flood event, from initial risk prediction through relief-camp distribution planning — the operational counterpart to Fig. 1. Authentication uses JSON Web Tokens with bcrypt password hashing and role-based access separating public, disaster-officer, healthcare-officer, and rescue-team roles. Machine-learning models are served as

independent microservices behind the integration layer, allowing a model (e.g., Module 4's Multi-Output Random Forest) to be retrained and redeployed without affecting the other modules.

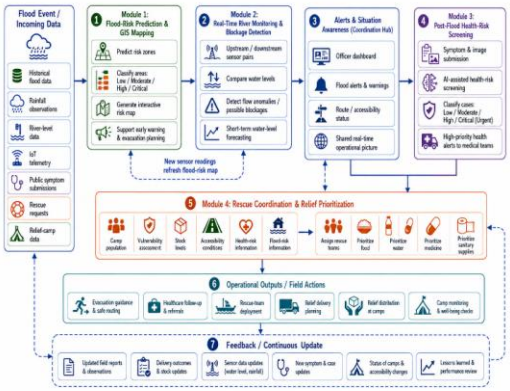


Fig. 2. Integrated cross-module disaster-management workflow.

V. IMPLEMENTATION

The platform follows a MERN-oriented web architecture with Python-based ML components and an IoT communication layer (Table II). The frontend uses React.js, TypeScript, Tailwind CSS, and Vite; Leaflet.js/React-Leaflet with OpenStreetMap and GeoJSON provide GIS visualization. The backend uses Node.js/Express.js REST APIs, with WebSocket/Socket.IO for time-sensitive alerts such as drainage anomalies, high-priority health cases, and rescue-status changes. ML components are implemented in Python with scikit-learn [14]: Random Forest for classification tasks [5] and Gradient Boosting for flood-risk and water-level forecasting [13], kept separate from the main application so models can be retrained independently. The IoT layer uses Arduino-based nodes with three ultrasonic sensors and a rainfall sensor, transmitting readings via Wi-Fi (MQTT/HTTP), consistent with established low-cost river-monitoring designs [7], [8]. MongoDB stores flood-risk, sensor, health-screening, and relief-camp records in a shared document-oriented store. Datasets are cleaned, encoded, and split into training/testing subsets, with cross-validation used to compare candidate models and reduce data leakage for temporally or spatially dependent data [5], [6].

The modular implementation allows the web application, IoT monitoring infrastructure, machine-learning services, geospatial visualization, and centralized database to be developed, tested, and scaled as interconnected but independently deployable components. Because each machine-learning service is exposed through the same integration layer as a stateless microservice, additional sensor nodes, relief camps, or user roles can be onboarded without redesigning the surrounding modules, and a single model can be retrained on newly collected Sri Lanka-specific data and redeployed with minimal disruption to the rest of the platform.

Layer	Technology
Frontend / Mapping	React.js, TypeScript, Tailwind CSS, Vite, Leaflet.js, GeoJSON
Backend APIs	Node.js, Express.js (REST), WebSocket/Socket.IO
ML services	Python, scikit-learn
IoT layer	Arduino C/C++, ultrasonic + rainfall sensors, Wi-Fi (HTTP/MQTT)
Data storage	MongoDB
Security	JWT authentication, bcrypt, role-based access control

TABLE II. TECHNOLOGY STACK SUMMARY

VI. RESULTS AND DISCUSSION

The proposed platform was evaluated at the module level, using measures appropriate to each module's function, and the complete prototype was further assessed for cross-module information exchange. The results were then considered together to determine how effectively the four components can support coordinated flood-disaster management, and to identify expected benefits and current system limitations.

A. Flood-Risk Prediction and GIS Mapping

The flood-prediction module was evaluated on historical rainfall, river-level, elevation, drainage, and land-use data using accuracy, precision, recall, F1-score, and AUC. Random Forest and Gradient Boosting each achieved accuracy and F1-score of 1.00 on a held-out test set, while Logistic Regression achieved 0.93 accuracy and 0.95 AUC (Table III). Given the small test set (n = 40), these results should be interpreted cautiously, and evaluation on a larger, more diverse dataset is planned to confirm generalization; the trend is nonetheless consistent with prior ensemble-learning studies [5], [6]. Predicted risk categories are displayed on the GIS interface and updated as new monitoring data arrive.

Model	Accuracy	F1-Score	AUC
Random Forest	1.00	1.00	1.00
Gradient Boosting	1.00	1.00	1.00
Logistic Regression	0.93	0.91	0.95

TABLE III. FLOOD-RISK PREDICTION MODEL COMPARISON

B. Drainage Monitoring, Blockage Detection, and Forecasting

The IoT monitoring module was evaluated using multi-point ultrasonic water-level measurements and rainfall observations. Forecasting was assessed with RMSE and MAE, while blockage/anomaly detection was assessed using correctly identified, false-positive, and false-negative alerts. The module achieved an RMSE of 0.15 m and MAE of 0.10 m, and correctly detected 47 of 50 anomaly events (94%), with 3 false positives and 2 false negatives. Comparing neighbouring monitoring points allowed the system to flag abnormal flow and approximate the blockage location, extending single-threshold IoT-monitoring approaches [7], [8].

C. Post-Flood Health-Risk Screening

The health-screening module was evaluated on its ability to classify cases into Low, Moderate, High, and Critical/Urgent categories, with particular attention to recall for High and Critical/Urgent cases, since missing an urgent case carries more risk than an over-cautious classification. The system achieved an overall F1-score of 0.92, with recall of 0.97 for the Critical/Urgent category (Table IV), supporting its intended triage role rather than independent diagnosis [9], [10].

Category	Precision	Recall	F1
Low	0.95	0.96	0.95
Moderate	0.90	0.89	0.89
High	0.89	0.92	0.90
Critical/Urgent	0.93	0.97	0.95

TABLE IV. POST-FLOOD HEALTH-RISK SCREENING PERFORMANCE

D. Rescue and Relief-Resource Prioritization

The Multi-Output Random Forest model was evaluated on relief-camp and rescue-related data, producing five outputs: overall camp priority and resource-specific priorities for food, water, medicine, and sanitary supplies [5], each assessed independently using accuracy and F1-score. Water Priority achieved the highest F1-score (0.96), Overall Camp Priority the lowest (0.92), with Food (0.95), Medicine (0.94), and Sanitary-Supply (0.93) between (Table V) — indicating that a shared camp-level feature set can support several related priority predictions. Outputs are further reviewed against humanitarian urgency, fairness, and Sphere minimum-assistance standards [11], [4], and accessibility data from Module 2 can flag unsafe routes before rescue or delivery teams are deployed.

Output	Accuracy	F1
Overall Camp Priority	0.92	0.92
Food Priority	0.95	0.95
Water Priority	0.96	0.96
Medicine Priority	0.94	0.94
Sanitary-Supply Priority	0.93	0.93

TABLE V. MULTI-OUTPUT RELIEF-PRIORITIZATION PERFORMANCE

E. Integrated System Evaluation and Discussion

Integration testing confirmed that outputs generated in one module could be consumed by another without manual re-entry: new river-level data from Module 2 update Module 1's risk assessment, flood-risk and accessibility information support Module 4 rescue and distribution decisions, and Module 3 health-risk information contributes to medical-support prioritization. A consolidated summary of the principal outcomes is presented in Table VI.

Module	Method	Result
Flood-Risk Prediction	RF / GBoost	1.00 (RF & GB)
Drainage Monitoring	Ultrasonic + GBoost	RMSE 0.15 m; 94%
Health-Risk Screening	Rules + ML	0.92 overall F1
Rescue & Relief	Multi-Output RF	0.92–0.96

TABLE VI. CONSOLIDATED PERFORMANCE SUMMARY

Compared with isolated disaster-management tools, the integrated platform provides a broader operational view combining environmental conditions, affected populations, health needs, rescue activity, and resource requirements, which may improve early response, resource prioritization, and situational awareness throughout the flood-disaster lifecycle. However, system reliability remains dependent on the quality and quantity of the data used for training and evaluation; further field validation using larger Sri Lanka-specific flood datasets, real IoT sensor deployments, healthcare records, and operational relief-camp data will therefore be required before the system can be considered suitable for large-scale deployment.

VII. CONCLUSION

This paper presented an integrated AI-, IoT-, and GIS-based flood-disaster management platform for Sri Lanka that connects flood-risk prediction, real-time drainage monitoring, post-flood health-risk screening, and rescue and relief prioritization through a shared integration layer. The central contribution is the coordinated exchange of information across the disaster-management workflow rather than four isolated tools, which is expected to improve situational awareness and reduce response delays. The modular architecture also allows individual components to be updated or replaced independently. Future work will focus on collecting Sri Lanka-specific datasets, conducting pilot IoT deployments, evaluating the health-screening component under appropriate data governance, testing relief prioritization with operational records, and exploring offline/edge processing and alternative alert channels for low-connectivity areas, with human oversight and Sphere minimum standards [4] remaining central to the decision-support process.

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